Computer Algorithm

Tutorial No. 1

1. What is an algorithm and give its characteristics.
2. Explain different asymptotic Notations.
3. Explain performance analysis and performance measurement.
4. Explain Tower Of Hanoi problem in detail and analyze its complexity.
5. Compare the complexities of n, logn,nlogn,n^2,n^3,2^n.
6. The factorial function n! Has value 1 where n<=1 and value n*(n 1) when n>1,write both recursive and iterative algorithm t compute n!.
7. The Fibonacci numbers are defined as f0=0,f1=1 and fi=fi-1+fi-2 for i>1. Write both recursive and iterative algorithm to compute fi.

Tutorial No.2

1. Explain divide and conquer algorithm with recurrence relation.
2. Show that number of comparisons for finding maximum and minimum is 3n/2-2 where n is the no. of elements.
3. Explain recursive binary search algorithm with example.
show that complexity of binary search is O(logn) for successful search
4. Explain merge sort algorithm with example and calculate its complexity.

. Explain quick sort algorithm with example and and show that its complexity is O(nlogn).

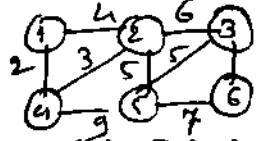
. Explain selection sort algorithm and show that its complexity is)(n^2). or write the algorithm to select nth smallest element in the list & compute it's complexity

Tutorial N0.3

. Write a short note on Greedy method.

. Find an optimal solution to knapsack instance n=6,c=20,w=(w1,w2,w3,w4,w5)=(2,3,5,7,1,5) and profit is(p1,p2,p3,p4,p5)=(12,5,15,7,6,18).

. Prove that if P1/W1 >=P2/W2 >= ...>= Pn/Wn then Greedy method generates an optimal solution to the given instance of knapsack problem.



. Define Spanning tree with suitable example and explain Prim's algorithm to find Minimum Cost Spanning Tree.

. Explain Kruskal's algorithm and compute Minimum Cost Spanning tree with example.

Solve job sequencing with deadline problem using greedy approach for following problem instances.
n=5, (P1, P2, P3, P4, P5) & (d1, d2, d3, d4, d5)
45 15 55 65 75 1, 3, 2, 1, 2.

Tutorial N0.4

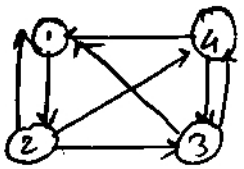
1. Write a short note on Optimal Storage on Tapes.

2. Obtain a Huffman code for message (m1,m2,....,mn) with relative frequencies (4,5,7,8,10,12,20). Draw decode tree for this set of codes.

3. Find an optimal binary merge pattern for 10 files whose lengths are 28,32,12,5,84,53,91,35,3.

4. Write a note on single source shortest path algorithm.

Tutorial No.5



$$\begin{bmatrix} 0 & 5 & \infty & \infty \\ 50 & 0 & 15 & 5 \\ 30 & \infty & 0 & 15 \\ 15 & \infty & 5 & 0 \end{bmatrix}$$

1. What is a multistage graph?

2. Explain all pair shortest path using dynamic programming.

3. Obtain Optimal binary search tree for identifiers set A1,A2,A3=(do,if,while) with probability p1=0.5,p2=0.1,p3=0.05 and q0=0.15,q1=0.1,q2=0.05,q3=0.05.

4. Explain 0/1 knapsack problem with example in dynamic programming.

5. Explain with example Travelling Sales Person problem.

6. Solve the following instances of reliability design problem with 3 stages and cost of system is 105.Cost of devices of stage 1 is 30,stages 2 is 15 and for stage 3 is 20.Reliabilities of stages are 0.9,0.8 and 0.5 respectively.

Tutorial N0.6

1. Explain binary tree traversal techniques with example.

2.Explain searching techniques for graph.

3.How can be a non connected graph is converted into connected graph. Define an articulation point how non connected graph can be converted to bi connected component.

4.Write a short note on

1. Game Tree
2. And /OR Graph

Tutorial N0.7

1. Draw and Explain a permutation tree for 4 queen problem.
   Explain 4queen Problem (write algorithm to test no 2 queens are placed in the same diagonal, row & column.
2. Explain Sum Of Subset problem using backtracking.
3. Explain Hamilton cycle with suitable example.
4. Explain Graph Coloring problem using backtracking.
5. Expalin the following terms with suitable example.
   1. Live Node
   2. E Node
   3. Dead Node
   4. Bounding Function.

Tutorial N0.8

1. With neat diagram explain relationship between P,NP,NP-hard ,NP-Complete.
2. List and Explain NP-hard problems. → graph
   clique Decision Problem
   Node cover decision Problem
   Directed Hamiltonian cycle.
   chromatic number Decision Problem.
3. Define the following terms

1. Deterministic Algorithm

2. Non –Deterministic Algorithm

Tutorial N0.9

1. Describe and give its example of prefix computational model with PRAM.

2. List and Explain variants of PRAM.

3. Explain 3D Butterfly network for hypercube.

4. Explain Odd Even merges on mesh with neat diagram.

5. Broadcasting with mesh & hypercube

6. Prefix sum computation in hypercube.

7. Prefix sum computation with mesh & hyperbe

8. what is a determinstic list ranking problem in PRAM, Explain with example.

9. Explain PRAM computational model

10. Explain 'packet routing in square & mesh & linear array.

11. Data concentration with mesh & hypercube.